

HealthConnect Analytics Testing & Refinement Report

AnalystLab Africa Experience Lab -Week 7

1. Week 6 to Week 7 Transition

In Week 6, I carried out deeper analysis on the HealthConnect appointment data and validated some of the main findings from my earlier work. Booking lead time, previous no-show history and reminder status remained important areas in understanding appointment no-shows.

I also improved my Power BI report by adding a separate page for the advanced analysis and collaborated with the Data Science track to validate the more detailed booking lead-time grouping.

For Week 7, I am building directly on that work by testing whether the main KPIs and analytical findings are accurate and consistent, checking whether the Power BI dashboard behaves correctly when filters are applied, and testing some of the Week 6 findings together rather than looking at them separately.

One of the main areas I decided to test further was the relationship between booking lead time and reminder status. In Week 6, these were analysed separately, so I wanted to see whether the booking lead-time pattern still remained when reminder status was considered at the same time.

2. Week 7 Testing Approach

For Week 7, I focused on testing the accuracy and consistency of the analytical work I had already completed rather than starting a new analysis.

I began by going back to the underlying dataset in Excel and recalculating some of the main KPIs and findings from Week 6. I checked the overall no-show rate, average booking lead time, reminder-status no-show rates and previous no-show history.

I then tested the Power BI dashboard to make sure that the values shown on the dashboard agreed with the underlying data when filters were applied. I tested individual filters as well as combined filters.

After validating the existing results, I carried out a new Week 7 test by combining booking lead time with reminder status. The purpose was to see whether the booking lead-time pattern I found in Week 6 would still remain when reminder status was considered at the same time.

For each test, I recorded the expected and actual result to see if the test passed or failed, any issue identified, the action taken and the result after re-testing. This helped me keep a clear record of what was tested and what was confirmed or improved.

3. KPI Validation

As part of my Week 7 testing, I went back to the underlying data in Excel to check whether some of the main KPIs and analytical findings from Week 6 were still accurate.

The first KPI I tested was the overall no-show rate. The dataset contained 5,000 appointments, with 2,423 recorded as no-shows. When I recalculated the rate, I got 48.46%, which matched the value already shown on my Power BI dashboard.

I also rechecked the average booking lead time by appointment outcome. Attended appointments had an average booking lead time of 24.52 days, while no-show appointments had an average of 34.53 days. These results also matched my Week 6

2 findings.

The reminder-status result was tested again by comparing attended appointments with no-shows while excluding cancellations. Appointments where no reminder was sent had a no-show rate of 54.63%, while appointments where a reminder was sent had a lower no-show rate of 49.86%.

I also validated the previous no-show history analysis. The no-show rate was 46.30% for patients with no previous no-show, 55.87% for those with one previous no-show, 62.05% for those with two, and 70.33% for patients with three or more previous no-shows.

All four tests matched the results I had previously reported, so the main Week 6 KPIs and findings were successfully validated.

The Week 7 assignment requires important KPI calculations and analytical findings to be checked against the underlying data, rather than simply carried forward from Week 6.

4. Dashboard Testing and Validation

After validating the calculations in Excel, I tested the Power BI dashboard to make sure that the figures shown on the dashboard agreed with the underlying data when filters were applied.

One of the filters I tested was Appointment Type. When I selected Follow-up appointments, Excel showed a total of 1,421 appointments and a no-show rate of 51.23%. The Power BI dashboard returned the same values.

When I selected Female patients, Excel showed 2,488 appointments and a no-show rate of 48.43%, and Power BI returned the same values.

For the combined test, I selected Follow-up appointments and Female patients. Excel returned 708 appointments with a no-show rate of 50.99%. Power BI also returned 708 appointments and a no-show rate of 50.99%.

After clearing the filters, the dashboard returned to the original 5,000 appointments and overall no-show rate of 48.46%.

These tests confirmed that the dashboard filters and calculations were responding correctly and that the filtered Power BI results agreed with the underlying data.

5. Analytical Testing: Booking Lead Time and Reminder Status

One of the main things I wanted to test further in Week 7 was the relationship between booking lead time and reminder status.

In Week 6, I had analysed booking lead time and reminder status separately. For Week 7, I decided to combine them to see whether the booking lead-time pattern would still remain when reminder status was considered at the same time.

The results showed:

- 1) 0–7 days: 31.25% no-show without reminder and 28.83% with reminder
- 2) 8–14 days: 37.33% without reminder and 34.43% with reminder
- 3) 15–30 days: 49.72% without reminder and 43.91% with reminder
- 4) 31–60 days: 67.63% without reminder and 62.58% with reminder

The results confirmed that the booking lead-time pattern still remained. In both reminder groups, the no-show rate increased as booking lead time became longer.

The analysis also showed that the no-show rate was lower where a reminder was sent in every booking lead-time group.

The difference between appointments with and without reminders was about 2.42 percentage points for 0–7 days, 2.90 points for 8–14 days, 5.81 points for 15–30 days and 5.05 points for 31–60 days.

This gave me a clearer picture of how booking lead time and reminder status behaved together.

6. Dashboard Refinement

In Week 6, booking lead time and reminder status were presented separately. Although the findings were useful, it was not easy to see how both factors behaved when considered together.

Based on the Week 7 testing, I added a new page called Week 7 Validation & Refinement.

I added a matrix showing the exact no-show rates for each booking lead-time group by reminder status. I also created a clustered column chart titled No-Show Rate by Booking Lead Time and Reminder Status.

The matrix makes it easier to see the exact percentages, while the chart makes the overall pattern easier to compare visually.

After creating the new visuals, I checked the figures again and confirmed that the values shown in the matrix and chart were consistent.

This refinement made the interaction between booking lead time and reminder status clearer and improved the business interpretation of the dashboard.

7. Validated Findings and Updated Business Insights

The Week 7 testing helped confirm that the main findings from Week 6 were still reliable.

The overall no-show rate remained 48.46%, and the average booking lead time remained higher for no-shows than for attended appointments.

Previous no-show history also continued to show a clear pattern, with patients who had missed appointments before recording higher no-show rates.

The new Week 7 analysis strengthened the booking lead-time finding. Even after reminder status was considered, no-show rates continued to increase as booking lead time became longer.

Another useful finding was that appointments where reminders were sent had lower no-show rates across every booking lead-time group.

The difference was larger among appointments booked 15 days or more in advance. This gave me a clearer descriptive view of how booking lead time and reminder status relate to no-show rates when considered together.

However, this is still an observational analysis. It shows an association between reminders and lower no-show rates, but it does not prove that the reminders alone caused the difference.

8. Updated Recommendations

Based on the Week 7 testing, I would still recommend that HealthConnect continue using appointment reminders. However, rather than treating all appointments the same way, the reminder approach could become more targeted.

Patients whose appointments are booked 15 days or more in advance may need more attention because these groups recorded higher no-show rates and also showed a larger difference between appointments with and without reminders.

HealthConnect could consider giving more reminder attention to appointments booked far in advance, such as by sending an earlier reminder and another reminder closer to the appointment date.

Previous no-show history should also continue to be considered when deciding which patients may need additional follow-up.

At the same time, the results should be used carefully. The analysis shows relationships within the available dataset and should support decision-making rather than be treated as proof that any one factor directly causes a patient to miss an appointment.

9. Cross-Track Testing and Refinement

For my Week 7 cross-track activity, I continued working with the Data Science track.

In Week 6, our collaboration focused on refining and validating the booking lead-time grouping. For Week 7, I shared the new result from my combined booking lead-time and reminder-status analysis.

I provided the following results:

1. 0–7 days: 31.25% without reminder and 28.83% with reminder
2. 8–14 days: 37.33% without reminder and 34.43% with reminder
3. 15–30 days: 49.72% without reminder and 43.91% with reminder
4. 31–60 days: 67.63% without reminder and 62.58% with reminder

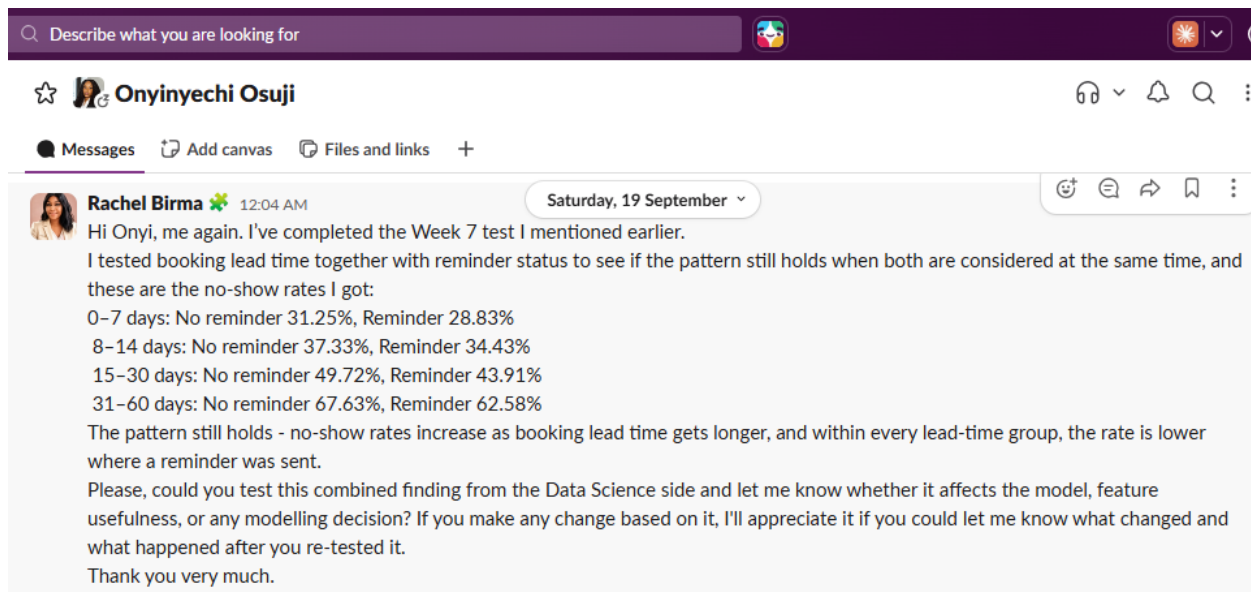


Figure 1: Cross-track testing request shared with the Data Science collaborator.

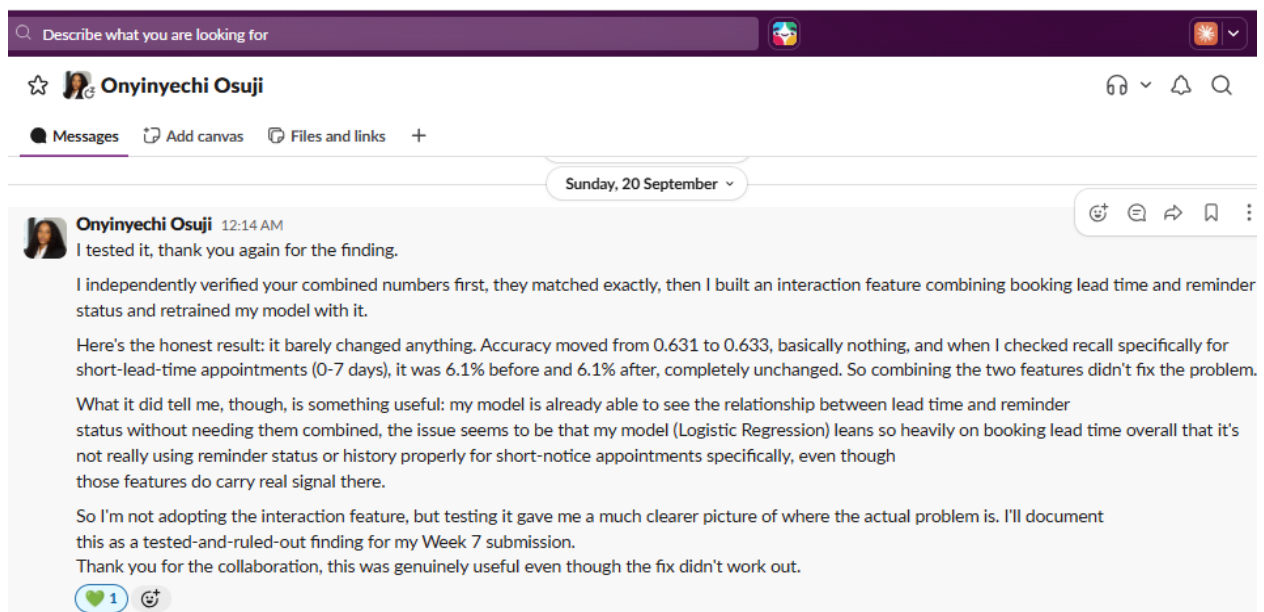


Figure 2: Data Science validation and re-testing of the combined booking lead-time and reminder-status finding.

The screenshots above provide evidence of the cross-track testing process. The Data Science collaborator independently verified the figures before testing the interaction feature in the Logistic Regression model.

The finding I shared was that booking lead time continued to show the same pattern after reminder status was introduced, and reminder use was associated with a lower no-show rate across all four booking lead-time groups.

I asked my Data Science collaborator to test whether this combined finding affected feature usefulness, model behaviour or any modelling decision, and to share what happened if any refinement was made and tested again.

My Data Science collaborator independently verified the figures I shared and confirmed that they matched her calculations. She then created an interaction feature combining booking lead time and reminder status and retrained the Logistic Regression model to test whether the combined feature improved the model.

The result showed only a very small change in accuracy, from 0.631 to 0.633. Recall for short-lead-time appointments (0–7 days) remained unchanged at 6.1%. Based on this result, the interaction feature was not adopted.

Although the feature did not improve the model, the test was still useful because it showed that combining booking lead time and reminder status did not solve the short-lead-time prediction issue. This completed the cross-track testing cycle by taking the analytical finding from Data Analytics, testing it from the Data Science side and reviewing the result before deciding whether to adopt the refinement.

10.Limitations and Remaining Analytical Issues

Although the Week 7 testing confirmed several important findings, there are still some limitations to consider.

Some analyses, including the reminder-status and previous no-show comparisons, excluded cancelled appointments so that attended appointments could be compared directly with no-shows. These percentages therefore use a different denominator from the overall no-show rate of 48.46%, which includes all 5,000 appointments.

The group with three or more previous no-shows is also relatively small compared with the other groups, so the 70.33% no-show rate should still be interpreted with caution.

The analysis also identifies relationships rather than causation. For example, appointments with reminders recorded lower no-show rates, but this analysis alone cannot prove that reminders caused the reduction.

Dashboard filter testing was carried out on selected individual and combined filter scenarios rather than every possible filter combination.

The Data Science test showed that the interaction feature produced only a small change in accuracy and did not improve recall for short-lead-time appointments. The feature was therefore not adopted. This result also shows that further modelling refinement may be needed before the analytical findings can be translated into stronger predictive performance.

11. Week 8 Readiness and Recommendations

The Week 7 testing has given me more confidence that the main analytical outputs are accurate and consistent.

Before Week 8, I will keep the validated KPI definitions, the refined Week 7 visualisation and the updated recommendation around booking lead time and reminder support.

I have incorporated the Data Science testing result into my Week 7 documentation, including the decision not to adopt the interaction feature.

For final integration, I will carry out one more review of the dashboard, calculations and supporting documentation to make sure the figures remain consistent and that the final presentation communicates the most important findings clearly.

My main focus for Week 8 will therefore be final integration and presentation rather than carrying out another complete analysis.

12. Week 7 Project Summary

My main focus in Week 7 was to test and refine the analytical work I had already completed rather than start another analysis.

I validated the overall no-show rate, average booking lead time, reminder-status result and previous no-show history against the underlying data. All of these tests matched my Week 6 results.

I also tested the Power BI dashboard filters against Excel and confirmed that the filtered dashboard values agreed with the underlying data.

For the deeper Week 7 analysis, I tested booking lead time together with reminder status. The booking lead-time pattern remained clear, with no-show rates increasing as appointments were booked further in advance. I also found that no-show rates were lower where reminders were sent across all four booking lead-time groups.

Based on this result, I refined my Power BI report by adding a Week 7 Validation & Refinement page with a matrix and clustered column chart showing booking lead time and reminder status together.

I also continued my cross-track collaboration with Data Science by sharing the validated combined finding for further testing from the modelling side. The finding was independently verified and an interaction feature was tested in the Logistic Regression model. The feature produced only a small change in accuracy and did not improve short-lead-time recall, so it was not adopted.

The main lesson from this week was that testing is not only about finding errors. It can also help confirm that earlier findings are reliable, show where an analysis can be improved and make the final recommendations more useful.

My focus before Week 8 is to make sure all final figures and visuals are consistent and prepare the analytical work for final integration and presentation.